Gradient Descent on a Riemannian Manifold

The Euclidean Picture

Let

$$f : \mathbb{R}^n \to \mathbb{R}$$

be smooth. In Euclidean space, the gradient $\nabla f(x)$ is characterized by

$$Df(x)[v] = \nabla f(x) \cdot v$$

for every direction $v \in \mathbb{R}^n$.

Here $Df(x)[v]$ is the directional derivative of f at x in the direction v. The dot product is the Euclidean metric. Thus the gradient is not merely "the list of partial derivatives." It is the vector that represents the linear map $Df(x)$ through the chosen inner product.

Differential Versus Gradient

On a smooth manifold M, the derivative of a smooth function

$$f : M \to \mathbb{R}$$

at a point $p \in M$ is naturally a covector

$$df_p : T_pM \to \mathbb{R}.$$

It eats a tangent vector $X_p \in T_pM$ and returns the directional derivative

$$df_p(X_p) = X_p[f].$$

So the differential df_p lives in the cotangent space T_p^*M, not in the tangent space T_pM. To turn this covector into a vector, we need extra geometric structure.

The Metric Converts Covectors Into Vectors

A Riemannian metric assigns to each point $p \in M$ an inner product

$$g_p : T_pM \times T_pM \to \mathbb{R}.$$

The Riemannian gradient of f is the unique tangent vector

$$\operatorname{grad}_g f(p) \in T_pM$$

such that

$$df_p(X_p) = g_p(\operatorname{grad}_g f(p), X_p)$$

for every tangent vector $X_p \in T_pM$.

This is the precise meaning of the phrase:

> The gradient is the vector that represents the directional derivative through the metric.

The directional derivative is the covector df_p. The metric g_p converts that covector into the vector $\operatorname{grad}_g f(p)$.

Coordinate Formula

Let $(x^1, \dots, x^n)$ be local coordinates. Write the metric as

$$g = g_{ij}\, dx^i \otimes dx^j.$$

The differential is

$$df = \frac{\partial f}{\partial x^i}\, dx^i.$$

If

$$\operatorname{grad}_g f = (\operatorname{grad}_g f)^i \frac{\partial}{\partial x^i},$$

then the defining identity

$$df(X) = g(\operatorname{grad}_g f, X)$$

implies

$$\frac{\partial f}{\partial x^j} = g_{ij} (\operatorname{grad}_g f)^i.$$

Multiplying by the inverse metric g^{kj} gives

$$(\operatorname{grad}_g f)^k = g^{kj} \frac{\partial f}{\partial x^j}.$$

Thus

$$\operatorname{grad}_g f = g^{ij} \frac{\partial f}{\partial x^j} \frac{\partial}{\partial x^i}.$$

When $g_{ij} = \delta_{ij}$, this reduces to the usual Euclidean gradient.

Gradient Flow

The Riemannian gradient flow of f is the curve $p_t \in M$ solving

$$\dot{p}_t = -\operatorname{grad}_g f(p_t).$$

This is the intrinsic version of gradient descent. The negative sign means that the curve moves in the direction of steepest decrease as measured by the metric g.

Indeed, along a solution,

$$\frac{d}{dt} f(p_t) = df_{p_t}(\dot{p}_t).$$

Using the defining property of the gradient,

$$df_{p_t}(\dot{p}_t) = g_{p_t}(\operatorname{grad}_g f(p_t), \dot{p}_t).$$

Substituting

$$\dot{p}_t = -\operatorname{grad}_g f(p_t)$$

gives

$$\frac{d}{dt} f(p_t) = -g_{p_t}(\operatorname{grad}_g f(p_t), \operatorname{grad}_g f(p_t)) \le 0.$$

So the objective decreases monotonically.

Why This Matters for Wasserstein Geometry

The Wasserstein calculation follows the same pattern.

On a finite-dimensional Riemannian manifold:

$$df_p(X_p) = g_p(\operatorname{grad}_g f(p), X_p).$$

In Wasserstein geometry, a tangent direction is represented by a velocity field v, and the metric is the kinetic-energy inner product

$$\langle u, v \rangle_\rho = \int_{\mathbb{R}^d} u(x) \cdot v(x)\, \rho(x)\, dx.$$

Thus the Wasserstein gradient is identified by writing the directional derivative in the form

$$D\mathcal{F}(\rho)[v] = \langle \operatorname{grad}_{W_2} \mathcal{F}(\rho), v \rangle_\rho.$$

This is why, once we obtain

$$D\mathcal{F}(\rho)[v] = \int_{\mathbb{R}^d} \nabla \left(\frac{\delta \mathcal{F}}{\delta \rho} \right) \cdot v\, \rho\, dx,$$

we identify the velocity-field representative of the Wasserstein gradient as

$$\operatorname{grad}_{W_2} \mathcal{F}(\rho) = \nabla \left(\frac{\delta \mathcal{F}}{\delta \rho} \right).$$